

Qian Ouyang

Machine learning for materials and molecular simulation.

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Education

Xi'an Jiaotong University

Fourth-year undergraduate | Artificial Intelligence
Artificial Intelligence Experimental Class
2023 - 2027 (expected)

Academic experience

Research Assistant

University of Notre Dame
Prof. Tengfei Luo's group | Department of Aerospace and Mechanical Engineering

October 2025 - Present

Summer Research Participant (iSURE)

University of Notre Dame
International Summer Undergraduate Research Experience (iSURE).

July - September 2026

Visiting Student

University of Notre Dame

August - December 2025

Research interests

AI for Materials; Molecular Dynamics & Machine Learning Potentials; HPC Optimization of ML Interatomic Potentials; Coarse-Grained Molecular Modeling; Representation Learning / JEPA; Polymer Property Prediction.

Selected research

Molecular dynamics representation learning

From CG JEPA and encoder comparisons to ongoing MACE-based atomistic representation learning, evaluated through physical readouts and temporal controls.

MACE JEPA: representations for molecular dynamics

Ongoing research
Investigate MACE-based molecular representations using position-velocity adapters, temporal JEPA controls, physical readouts, and hydrogen-bond exchange propensity targets. Completed water pilots show task-dependent effects.

CG JEPA: polymer representations and physical prediction

Exploratory phase completed
Investigated JEPA-style polymer representations through controlled ablations, geometric interfaces, and physical-readout evaluation. Temporal variants did not surpass strong linear baselines in the tested setting.

Selected research (continued)

MLIP material properties and efficiency

A connected line of polymer property workflows and resource evaluation: UMA / FAIR-Chem exploration, preliminary MACE density screening, and a bounded GPU engineering pilot.

Preliminary polymer density screening with MACE

Preliminary screening | 17 September 2026

Analyzed preliminary MACE density estimates for three polymers, including sampling behavior and reference-data matching. One packing and 25 ps sampling per polymer; experimental conditions remain incompletely matched.

GPU resource trade-offs in MACE molecular dynamics

Engineering pilot | September 2026

Evaluated runtime, sampled per-GPU memory, and GPU-hour trade-offs for a fixed MACE polymer simulation workload. Engineering pilot with one short run per configuration; not a general scaling claim.

Polymer elastic-response workflows with UMA / FAIR-Chem

Resource-limited exploration

Explored UMA / FAIR-Chem polymer calculations with LAMMPS. GPU-memory constraints limited progress before validated property results and motivated my interest in MLIP efficiency.

Technical skills

Molecular simulation & MLIPs: Experience with UMA/FAIR-Chem and MACE for molecular simulation.

High-performance computing: Experience using Notre Dame's Center for Research Computing (CRC) for research computing.

Programming languages: C++, Python